EXPLORATORY DATA ANALYSIS

Chapter 16

Outlier and Anomaly Detection

Finding unusual observations without confusing rarity with error

Course: From Raw Data to Understanding

Focus: Univariate outliers • Robust scores • Multivariate anomalies • Context

A mathematically grounded, practical chapter using Python, NumPy, Pandas, Matplotlib's object-oriented API, and Seaborn.

Learning objectives

- Distinguish an outlier, an anomaly, an error, and a rare but valid observation.
- Apply IQR fences, modified z-scores, and standard z-scores with appropriate caveats.
- Understand why univariate methods can miss multivariate anomalies.
- Recognize global, contextual, and collective anomalies.
- Use robust summaries and visual diagnostics to investigate flagged observations.
- Understand the assumptions and limitations of distance- and density-based detection methods.
- Design a reproducible flagging workflow that preserves original observations and records decisions.

How this chapter fits the course

Earlier chapters taught us to describe distributions, compare groups, and investigate relationships. Unusual observations can distort summaries, reveal data-quality problems, identify important subpopulations, or expose genuinely rare events. This chapter develops methods for finding candidates for investigation—not automatic rules for deleting data.

Review of earlier chapters (excluding Chapter 15)

- Chapters 1–3: EDA is iterative; define the observational unit, understand the data-generating process, inspect metadata, and assess missingness, validity, consistency, and uniqueness.
- Chapters 4–6: distinguish numerical, categorical, and temporal variables; interpret units, granularity, category levels, and time ordering correctly.
- Chapters 7–9: use robust descriptive statistics, quantiles, IQR, skewness, kurtosis, and distribution shape to understand what 'unusual' means relative to a variable's distribution.
- Chapters 10–11: use histograms, ECDFs, boxplots, scatterplots, and conditional views to inspect tails, clusters, unusual pairs, and nonlinear patterns.
- Chapter 12: use multivariate displays and facets; visual encodings and scaling can reveal or hide unusual combinations.
- Chapters 13–14: understand association, covariance, and dependence; a point may be unusual because of its relationship with other variables, even if each individual value looks ordinary.

Review of the immediately previous chapter: Chapter 15 — Grouped Analysis

GroupBy, crosstabs, and pivot tables compare subsets of observations. Correct interpretation depends on the observational unit, group size, denominators, missing-key policy, and aggregation level. Pooled summaries can hide group-specific patterns, and Simpson's paradox warns that composition can change an aggregate comparison.

Chapter 15 assignment/project: expected-reasoning guide

- Aggregation produces group-level summaries; transformation returns row-aligned group values; filtering retains or removes whole groups.
- Count and size differ when selected values are missing.
- Row, column, and overall crosstab normalization answer different denominator questions.
- A missing pivot cell is not automatically a measured zero.
- Unweighted averages of group means can differ from the overall mean; group sizes or exposures matter.

This is an expected-reasoning guide, not a grading of submitted work.

Preview of this chapter

We will begin with the meaning of unusualness, then develop univariate detection methods before examining multivariate and contextual anomalies. We will finish with a workflow for flagging, validating, documenting, and communicating suspicious observations.

1. What is an outlier?

An outlier is an observation that lies unusually far from the bulk of a dataset under some chosen representation or model. The definition is relative: unusual compared with which population, variable, time period, subgroup, and distance measure?

An anomaly is often used more broadly for an observation or pattern that does not conform to expected behavior. In practice the terms overlap, but 'anomaly' often includes multivariate, temporal, contextual, or collective patterns.

1.1 Four ideas that must not be confused

- **Outlier:** statistically unusual under a specified distributional or geometric criterion.
- **Anomaly:** unusual relative to a model of expected behavior, possibly involving context or several variables.
- **Error:** an incorrect value caused by measurement, entry, processing, or system failure.
- **Rare valid observation:** uncommon but genuine behavior, such as a very large legitimate transaction.

> An unusual value is not proof of an error. A high salary may be valid; a negative age may be invalid; a sudden increase in server requests may be a genuine attack or a scheduled event. The data-generating context decides what deserves investigation.

1.2 Why detection matters

- Data quality: impossible values, unit mistakes, duplicated records, sensor failures.
- Analytical distortion: a few extreme observations can strongly affect means, variances, and correlations.
- Discovery: rare but important customer segments, failures, fraud patterns, or scientific events.
- Operational monitoring: unusual behavior may indicate an emerging incident.
- Modeling readiness: downstream methods may be sensitive to unusual observations, but handling belongs to a separate, justified preprocessing decision.

2. A taxonomy of anomalies

2.1 Global anomalies

A global anomaly is unusual relative to the full reference population. For example, a transaction of an exceptionally large amount may be rare across all transactions. Global detection can fail when populations contain legitimate subgroups with different scales.

2.2 Contextual anomalies

A contextual anomaly is unusual only under a particular context. A temperature of 28°C may be ordinary in one season and unusual in another. Context can include location, time, customer segment, device type, or operating conditions.

2.3 Collective anomalies

A collective anomaly is a group or sequence of observations that is unusual together, even when individual observations are not. Examples include an unusual sequence of login attempts or a sustained sensor pattern. Detecting these requires preserving ordering and group structure.

2.4 Point, contextual, and collective examples

Type	Illustrative example	Why a simple threshold may fail
Global	One transaction far above the overall range	A legitimate wholesale customer may have larger orders.
Contextual	Normal weekday traffic at an unusual hour	Traffic depends on day, season, and scheduled jobs.
Collective	A long sequence of individually normal readings	The sequence pattern, not any single value, is unusual.

3. Univariate detection: investigate one variable at a time

Univariate methods flag observations based on one variable's distribution. They are useful for screening, but they cannot identify every unusual combination of variables.

3.1 Visual inspection first

- Histogram: inspect tail length, gaps, spikes, and isolated bins.
- ECDF: see how much probability mass lies beyond a candidate value.
- Boxplot: use quartiles and IQR as a compact summary, while remembering that whiskers are not hard limits.
- Dot/strip plot: inspect individual values when the dataset is small enough.
- Log-scale display: useful for highly skewed positive values, but state clearly that the axis is transformed.

3.2 The IQR rule and Tukey fences

Let Q1 and Q3 be the first and third quartiles. The interquartile range is IQR = Q3 – Q1. The conventional inner fences are:

> **Lower fence = Q1 – 1.5 × IQR**
> **Upper fence = Q3 + 1.5 × IQR**

Values outside these fences are often flagged as potential outliers. The 1.5 multiplier is a heuristic, not a universal probability threshold. The fences are especially useful as a robust exploratory screen, but can over-flag naturally skewed distributions or valid heavy tails.

Outer fences at 3 × IQR are sometimes used to distinguish more extreme candidates. Neither inner nor outer fences establish that a value is wrong.

3.3 IQR method in Pandas

```
def iqr_outlier_flags(series, multiplier=1.5):
    q1 = series.quantile(0.25)
    q3 = series.quantile(0.75)
    iqr = q3 - q1
    lower = q1 - multiplier * iqr
    upper = q3 + multiplier * iqr

    flags = (series < lower) | (series > upper)
    return pd.DataFrame({
        "value": series,
        "lower_fence": lower,
        "upper_fence": upper,
        "is_flagged": flags
    })

result = iqr_outlier_flags(df["amount"])
```

3.4 Modified z-score using the median absolute deviation

The median absolute deviation (MAD) is a robust measure of spread. First calculate MAD = median(|xi – median(X)|). A commonly used modified z-score is:

> **Mi = 0.6745 × (xi – median(X)) / MAD**

The constant 0.6745 makes the MAD-based scale comparable to the standard deviation under a normal distribution. A common screening threshold is |Mi| greater than 3.5. It is still a heuristic, not a universal decision boundary.

If MAD is zero—for example, when many observations share the median—the score is undefined or unstable. Do not divide by zero or silently manufacture a score. Use another method or investigate the distribution.

```
from scipy.stats import median_abs_deviation

def modified_z_scores(series):
    x = series.astype(float)
    median = x.median()
    mad = median_abs_deviation(
        x.dropna(),
        scale=1.0,
        nan_policy="omit"
    )
    if mad == 0 or np.isnan(mad):
        return pd.Series(np.nan, index=x.index)

    return 0.67448975 * (x - median) / mad

scores = modified_z_scores(df["amount"])
flags = scores.abs() > 3.5
```

3.5 Standard z-scores

The standard z-score is z = (x – mean) / standard deviation. It measures distance from the mean in standard-deviation units. It is intuitive, but the mean and standard deviation are themselves sensitive to extreme values. It can be useful for roughly symmetric, well-behaved distributions, but should not be treated as a universal outlier detector.

```
x = df["amount"]
z = (x - x.mean()) / x.std(ddof=1)
z_flags = z.abs() > 3
```

3.6 Comparing common univariate methods

Method	Center / scale	Useful when	Main caution
IQR fences	Quartiles / IQR	Robust screening; skew-aware only to a degree	Fence multiplier is heuristic.
Modified z	Median / MAD	Robust scale and extreme-value screening	MAD may be zero; threshold is heuristic.
Standard z	Mean / SD	Roughly symmetric distributions	Extreme values affect their own scores.
Visual review	Distribution context	Understanding shape and data-generating process	Requires judgment; plots can hide points.

4. Distribution shape changes what looks unusual

A symmetric normal distribution is not the default shape of real-world data. Revenue, waiting times, counts, and durations are often right-skewed, bounded, zero-inflated, or multimodal.

- In a strongly right-skewed distribution, an upper IQR fence may flag many valid large values.
- In a multimodal distribution, a global center and spread can be misleading because subgroups overlap.
- In a bounded variable, the possible range matters; a generic z-score ignores domain limits.
- In a zero-inflated variable, the mass at zero and the positive-value distribution may need separate inspection.
- In a discrete count variable, ties and gaps are expected; continuous-distribution assumptions may not fit.

Before choosing a threshold, inspect the distribution and ask whether a transformation or domain-specific scale would make the comparison more meaningful. A log view can help visualize orders of magnitude, but it does not itself determine whether a record is erroneous.

5. Why univariate methods miss multivariate anomalies

An observation can be ordinary on every individual variable but unusual in combination. Imagine a person whose height and weight are each common values, but the particular pairing is rare relative to the population. Or a machine with normal temperature and normal vibration separately, but an unusual combination of both.

5.1 Scatterplots and conditional structure

A scatterplot can reveal points far from the main relationship even when their marginal values are ordinary. Investigate the relationship, clusters, and relevant groups before declaring a point anomalous.

5.2 Mahalanobis distance

Mahalanobis distance measures distance from a multivariate center while accounting for covariance among variables. For a vector x, mean vector μ, and covariance matrix Σ:

$$D^2(x) = (x - \mu)^\top \Sigma^{-1} (x - \mu)$$

Unlike Euclidean distance, Mahalanobis distance accounts for scale and linear correlation. A point far along a naturally high-variance direction may not be unusual, while a point far across a low-variance direction may be.

5.3 Assumptions and limitations

- Classical estimates of μ and Σ are sensitive to outliers, so extreme points can distort the reference geometry.
- Highly correlated variables can make Σ singular or ill-conditioned.
- Distance thresholds are often motivated by a chi-square approximation under multivariate normality; that calibration may be poor for skewed, heavy-tailed, clustered, or small-sample data.
- Categorical variables and mixed data types need different representations or methods.
- High dimensionality makes distances less discriminative and covariance estimation harder.

```
from scipy.spatial.distance import mahalanobis

features = ["height", "weight", "age"]
X = df[features].dropna()

center = X.mean().to_numpy()
cov = X.cov().to_numpy()
inv_cov = np.linalg.pinv(cov)

distances = X.apply(
    lambda row: mahalanobis(
```

```
        row.to_numpy(), center, inv_cov
    ),
    axis=1
)
```

This is an illustrative classical calculation, not a complete production detector. The pseudoinverse helps with numerical singularity but does not solve poor statistical assumptions. For robust multivariate detection, consider robust location and covariance estimates and validate results against plots and domain knowledge.

6. Robust multivariate screening

Robust methods attempt to estimate the center and spread without allowing a small number of extreme observations to dominate. One established approach is the Minimum Covariance Determinant (MCD), which seeks a subset whose covariance gives a robust representation of the main data cloud under suitable conditions.

```
from sklearn.covariance import MinCovDet

features = ["height", "weight", "age"]
X = df[features].dropna()

mcd = MinCovDet(random_state=42).fit(X)
robust_distances = mcd.mahalanobis(X)

# These are squared robust Mahalanobis distances.
```

MCD is not a universal solution. It is most suitable for numerical data with a dominant approximately elliptical central structure. Multiple legitimate clusters, strong nonlinearity, categorical features, and high dimensionality require other approaches.

6.1 Local Outlier Factor (LOF)

LOF compares the local density around an observation with the local densities around its neighbors. A point in a much sparser neighborhood than its neighbors can receive a high outlier score. This can detect local anomalies that a single global distance misses.

```
from sklearn.neighbors import LocalOutlierFactor

features = ["feature_a", "feature_b"]
X = df[features].dropna()

lof = LocalOutlierFactor(
    n_neighbors=20,
    contamination="auto"
)
labels = lof.fit_predict(X)  # -1 indicates outlier; 1 indicates inlier
scores = lof.negative_outlier_factor_
```

LOF is sensitive to feature scaling, the number of neighbors, local density, and dataset structure. Standardize numerical features when their units differ, but choose scaling with the data meaning in mind. The contamination setting and returned labels should not be treated as ground truth.

6.2 Isolation Forest

Isolation Forest identifies observations that are easier to isolate through random partitioning. Anomalies often have shorter average path lengths in the isolation trees. It can be useful for larger numerical datasets and does not require a distance metric in the same way as LOF.

```
from sklearn.ensemble import IsolationForest

features = ["feature_a", "feature_b", "feature_c"]
X = df[features].dropna()

detector = IsolationForest(
    n_estimators=200,
    contamination="auto",
    random_state=42
)
labels = detector.fit_predict(X)  # -1 = flagged, 1 = not flagged
scores = detector.decision_function(X)
```

Isolation Forest depends on representation, sample size, randomization, and tuning. It can flag unusual but valid subgroups. Inspect flagged records and compare the method with simpler baselines. A model's anomaly label is a screening output, not a factual diagnosis.

6.3 Choosing a method

Method	Main idea	Typical strength	Important limitation
Mahalanobis	Global covariance-adjusted distance	Elliptical numerical data	Sensitive to center/covariance and distribution shape.
Robust covariance / MCD	Robust central location and covariance	Outliers around one main cloud	Not ideal for multiple clusters or high dimensions.

Method	Main idea	Typical strength	Important limitation
LOF	Local density comparison	Local anomalies and varying density	Sensitive to scaling and neighborhood size.
Isolation Forest	Random partition isolation	Useful general-purpose screening	Scores and labels depend on representation and tuning.

7. Contextual and temporal anomalies

A global threshold can be inappropriate when expected behavior changes with time, location, season, or operating conditions. Contextual detection compares an observation with an appropriate reference group or baseline.

7.1 Context-specific baselines

- Compare a store's sales with its own weekday and seasonal pattern, not only all stores pooled together.
- Compare a device reading with the expected range for that device and operating state.
- Compare a transaction with similar customer or product contexts when those distinctions are meaningful.
- Check whether context variables are reliable and available at the time the observation would be assessed.

7.2 Temporal anomalies

For time-ordered data, inspect trend, seasonality, missing intervals, sudden level shifts, spikes, and unusual runs. A single extreme point and a sustained change in process are different events. Preserve timestamps, time zones, entity identifiers, and ordering.

A rolling median and rolling MAD can provide a robust local baseline, but window size and seasonality matter. Avoid using future observations when the real task is prospective detection.

```
# Illustrative rolling robust baseline for one regularly sampled series.
series = df.sort_values("timestamp").set_index("timestamp")["value"]

window = 25
rolling_median = series.rolling(
    window=window, center=False, min_periods=10
).median()

absolute_deviation = (series - rolling_median).abs()
rolling_mad = absolute_deviation.rolling(
    window=window, center=False, min_periods=10
).median()

rolling_score = 0.67448975 * (
    series - rolling_median
) / rolling_mad.replace(0, np.nan)
```

This simple rolling score is a screening illustration. It does not automatically account for weekly seasonality, changing variance, irregular sampling, or multiple entities. A baseline should reflect the process and the intended detection time.

8. A practical detection workflow

- **1. Define the unit:** What does one row represent? Is it a person, event, transaction, device reading, or time window?
- **2. Define the reference:** Which population, subgroup, time period, and operating context establish what is normal?
- **3. Validate data quality:** Check impossible values, units, parsing, duplicates, and known system changes first.
- **4. Explore distributions:** Use robust summaries and plots before selecting a detector.
- **5. Choose a method:** Match univariate, multivariate, local, or temporal detection to the question.
- **6. Flag, do not delete:** Store scores, thresholds, method, and parameters alongside the original data.
- **7. Investigate:** Review raw records, neighboring observations, related variables, groups, and domain rules.
- **8. Classify:** Possible error, valid rare event, contextual anomaly, process change, or unresolved case.
- **9. Document:** Record evidence, decision, owner, and impact on later analysis.
- **10. Reassess:** Check whether conclusions change with and without flagged cases; do not hide sensitivity.

9. Python: a reusable univariate flagging report

```
def univariate_outlier_report(series):
    x = series.dropna()
    if x.empty:
```

```
        raise ValueError("No non-missing observations.")

    q1 = x.quantile(0.25)
    q3 = x.quantile(0.75)
    iqr = q3 - q1
    lower = q1 - 1.5 * iqr
    upper = q3 + 1.5 * iqr

    median = x.median()
    mad = (x - median).abs().median()

    if mad > 0:
        modified_z = 0.67448975 * (x - median) / mad
    else:
        modified_z = pd.Series(
            np.nan, index=x.index
        )

    report = pd.DataFrame({
        "value": x,
        "iqr_flag": (x < lower) | (x > upper),
        "modified_z": modified_z,
        "modified_z_flag": modified_z.abs() > 3.5
    })
    metadata = {
        "n": len(x),
        "q1": q1,
        "median": median,
        "q3": q3,
        "iqr": iqr,
        "lower_fence": lower,
        "upper_fence": upper,
        "mad": mad
    }
    return report, metadata
```

A useful report returns both observation-level flags and the parameters used to create them. For client work, also record the source column, unit, missingness, reference population, and the interpretation of each flag.

10. Visualizing flagged observations

```
import matplotlib.pyplot as plt
import seaborn as sns

report, metadata = univariate_outlier_report(df["amount"])
plot_data = df.loc[report.index].copy()
plot_data["iqr_flag"] = report["iqr_flag"]

fig, ax = plt.subplots(figsize=(8, 4.5))
sns.boxplot(data=plot_data, x="amount", ax=ax)
ax.set_title("Amount distribution and IQR screening context")
fig.tight_layout()

fig, ax = plt.subplots(figsize=(8, 4.5))
sns.scatterplot(
    data=plot_data,
    x="feature_x",
    y="feature_y",
    hue="iqr_flag",
    ax=ax,
    alpha=0.75
)
ax.set_title("Flagged observations in relationship context")
fig.tight_layout()
```

The second plot illustrates an important principle: a univariate flag should be examined in the context of other variables. For multivariate methods, visualize flagged points against the variables or components that help explain why they were flagged.

11. Common mistakes

- Calling every flagged observation an error or fraud case.
- Applying a single global threshold to heterogeneous groups or seasonal data.
- Using mean and standard deviation when extreme values strongly distort both.
- Treating 1.5 × IQR, |z| > 3, or |modified z| > 3.5 as universal laws.
- Failing to inspect skewness, multimodality, zero inflation, or domain constraints.
- Assuming univariate screening detects unusual combinations of otherwise ordinary values.
- Using Mahalanobis distance without checking covariance conditioning and distributional assumptions.
- Ignoring feature scaling for distance- and neighborhood-based methods.
- Interpreting Isolation Forest or LOF labels as ground truth.

- Deleting flagged rows before investigating or documenting the reason.
- Comparing rolling scores across entities without context-specific baselines.

12. Summary

- Outliers and anomalies are relative to a reference population, representation, and context.
- An unusual observation may be an error, a valid rare event, or a meaningful process change.
- IQR fences, modified z-scores, and standard z-scores are screening tools with different assumptions.
- Multivariate anomalies can exist even when every individual feature looks ordinary.
- Mahalanobis distance, robust covariance, LOF, and Isolation Forest detect different structures and have limitations.
- Contextual and temporal anomalies require appropriate baselines and preservation of ordering.
- The professional workflow is to flag, investigate, classify, document, and only then decide whether any downstream treatment is justified.

13. Assignment

Part A — Concepts

- Distinguish an outlier, anomaly, error, and rare valid observation with one example of each.
- Why can the IQR rule over-flag observations in a strongly right-skewed distribution?
- Explain the role of MAD in the modified z-score and what happens when MAD equals zero.
- Give an example of a multivariate anomaly whose individual feature values are ordinary.
- Explain global, contextual, and collective anomalies.
- Why should an anomaly detector's output be treated as a flag rather than a diagnosis?

Part B — Numerical calculation

For X = [10, 11, 11, 12, 12, 13, 14, 15, 60], calculate Q1, Q3, IQR, and the 1.5 × IQR fences using Pandas' default linear quantile interpolation. Identify which observations are flagged. Then explain why the value 60 requires investigation but is not automatically invalid.

Part C — Method comparison

Choose a numerical variable from a real dataset. Compare IQR flags, modified z-score flags, and standard z-score flags. Include a histogram or ECDF, describe the distribution shape, and explain why the methods agree or disagree.

Part D — Multivariate investigation

Choose at least two numerical variables. Find a case where a point appears ordinary on each marginal distribution but unusual in the scatterplot. Explain the relationship that makes it unusual. If appropriate, calculate Mahalanobis distance and discuss its assumptions.

Part E — Context

Describe a contextual or temporal anomaly in a domain you understand. Define the reference group or baseline that makes the observation unusual, and explain what additional evidence would be needed before classifying it.

14. Coding project — Outlier Investigation Toolkit

Goal: Extend the cumulative EDA master notebook with a reproducible workflow for detecting and investigating unusual observations.

- Choose a dataset with at least three numerical variables and a clearly defined observational unit.
- Perform data-quality checks before anomaly detection; record impossible values and suspicious units separately.
- Create distribution plots and robust summaries for selected variables.

- Implement IQR and modified z-score screening with explicit handling of missing values and zero MAD.
- Compare univariate flags and show how many records each method identifies.
- Investigate at least one multivariate relationship using a scatterplot and, where appropriate, a multivariate method.
- Discuss whether global or subgroup-specific baselines are more meaningful.
- Preserve original rows and add flag columns or a separate findings table; do not delete flagged cases.
- Create an investigation log with record identifier, method, score, reason for review, evidence, and status.
- Write a short conclusion distinguishing data errors, valid rare observations, and unresolved anomalies.

Deliverables: a reproducible notebook section, plots, a flagged-record table, method metadata, and a concise investigation report. The purpose is to create an auditable screening process—not an automatic cleaning script.

Preview of Next Chapter

Chapter 17 — Investigating Unusual Observations

Chapter 16 focused on detection: how to find observations that deserve attention. Chapter 17 will focus on investigation: tracing unusual records to their source, testing data-quality and domain explanations, comparing records with relevant peers, measuring their influence on descriptive conclusions, and documenting whether they should be retained, corrected, excluded, or treated separately. The decision will be evidence-based, not threshold-based.

Selected references and further reading

- Tukey, J. W. Exploratory Data Analysis. Addison-Wesley, 1977.
- Hampel, F. R. et al. Robust Statistics: The Approach Based on Influence Functions. Wiley, 1986.
- Rousseeuw, P. J., & Van Driessen, K. A fast algorithm for the minimum covariance determinant estimator. Technometrics, 1999.
- Breunig, M. M. et al. LOF: Identifying Density-Based Local Outliers. SIGMOD, 2000.
- Liu, F. T., Ting, K. M., & Zhou, Z.-H. Isolation Forest. ICDM, 2008.
- Scikit-learn documentation: covariance estimation, LocalOutlierFactor, and IsolationForest.
- SciPy documentation: median_abs_deviation and distance functions.

This chapter is educational material for exploratory data analysis. Statistical flags identify observations for review; they do not establish error, intent, fraud, or causation.